

EMBEDDED SYSTEMS PROJECT

SMART ENVIRONMENTAL MONITORING SYSTEM

USING STM32

Real-time monitoring of air quality, temperature & humidity using embedded technology

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INTRODUCTION

- ▶ Air pollution is one of the most serious environmental issues affecting human health and ecosystems worldwide. Harmful gases such as CO, CH₄, and LPG can cause severe health hazards when present beyond safe limits.
- ▶ Environmental factors like temperature and humidity also play a significant role in human comfort, industrial processes, and climate conditions.
- ▶ This project presents a smart monitoring system that continuously tracks air quality and environmental parameters, providing real-time data using sensors and a microcontroller.

OBJECTIVE & NEED

OBJECTIVES

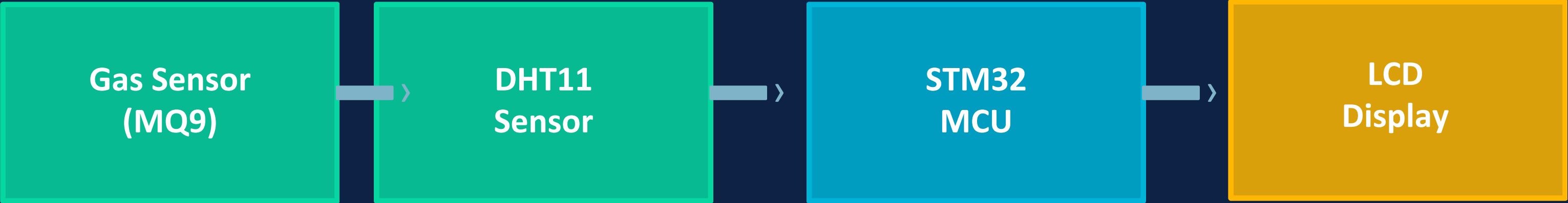
- 1 Develop a real-time environmental monitoring system
- 2 Detect harmful gases using gas sensors
- 3 Measure temperature and humidity accurately
- 4 Display data clearly for user awareness

WHY THIS SYSTEM?

- Increasing pollution levels in urban areas
- Lack of affordable monitoring devices
- Need for early detection of hazardous gases
- Helps prevent health risks and accidents

"Prevention is better than cure" — early detection saves lives.

SYSTEM OVERVIEW



Gas Detection	Env. Sensing	Processing	Display
MQ9 detects CO, CH ₄ & LPG via resistance change; ADC reads analog output	DHT11 measures temperature (0–50°C) & humidity (20–90% RH) digitally	STM32 processes sensor inputs, runs conversion logic and formats output	16×2 LCD (I2C) shows real-time sensor values in human-readable format

KEY FEATURES

Real-time Detection

Continuously monitors gas concentration with immediate ADC-based readings

Temp & Humidity

Factory-calibrated DHT11 provides accurate environmental measurements

Digital Display

Clear 16×2 I2C LCD output for instant user-readable data presentation

Compact & Portable

Breadboard-based modular design, easy to assemble and deploy anywhere

Low-Cost Design

Uses affordable, widely available components suitable for rapid prototyping

IoT-Ready

Expandable architecture — Wi-Fi/cloud integration possible as future scope

Designed for efficiency, simplicity, and scalability

HARDWARE COMPONENTS

CORE COMPONENTS

	STM32F103RB	32-bit ARM Cortex-M3 MCU @ 72 MHz
	MQ9 Gas Sensor	Detects CO, CH ₄ , LPG (analog out)
	DHT11 Sensor	Temperature (0–50°C) & Humidity
	16×2 LCD (I2C)	Real-time data display via I2C bus

SUPPORTING PARTS

- ◆ 10kΩ pull-up resistor (DHT11)
- ◆ Breadboard for prototyping
- ◆ Jumper wires for connections
- ◆ 5V USB / regulated power supply

Each component plays a critical role in data acquisition and display

STM32 MICROCONTROLLER

SPECIFICATIONS

Architecture:	32-bit ARM Cortex-M3
Clock Speed:	Up to 72 MHz
Flash Memory:	128 KB
SRAM:	20 KB
Package:	STM32F103RB

Acts as the central processing unit controlling the entire system

KEY FUNCTIONAL UNITS

ADC	Converts analog MQ9 signal to digital values for processing
GPIO	Handles single-wire DHT11 digital communication
I2C	Communicates with the 16×2 LCD display module
Timers	Manages sensor polling intervals and delays

MQ9 GAS SENSOR

DETECTS

CO Carbon Monoxide

CH₄ Methane

C₃H₈ LPG

WORKING PRINCIPLE

1

Sensing Material:

SnO₂ (tin dioxide) forms the active sensing layer on the sensor element

2

Resistance Change:

Gas concentration alters electrical resistance of the sensing material

3

Voltage Output:

Variable voltage proportional to gas concentration is produced at output

4

ADC Conversion:

STM32 ADC reads this analog voltage and converts it to a digital value

 **Important:** Requires preheating for stable readings. Calibration is needed for accurate gas concentration values.

DHT11 SENSOR

KEY FEATURES

Temperature Range	0°C – 50°C
Humidity Range	20% – 90% RH
Calibration	Factory calibrated (no manual calibration)
Protocol	Single-wire digital communication
Data Output	40-bit digital data frame

INTERNAL WORKING

Temperature

NTC thermistor measures ambient temperature accurately within the range

Humidity

Capacitive sensor detects moisture level changes in the surrounding air

On-chip MCU

Internal microcontroller handles sensor calibration and data encoding

Data Frame

Outputs a 40-bit frame: 8-bit humidity + 8-bit decimal + 8-bit temp + 8-bit decimal + 8-bit checksum

LCD DISPLAY & PIN CONFIGURATION

LCD (I2C INTERFACE)

- Reduces wiring complexity significantly
- Uses only 2 communication lines (SCL + SDA)
- Backpack module converts parallel to I2C

PIN MAPPING

PA0	MQ9 Analog Input
PA1	DHT11 Digital Input
PB6	I2C SCL (LCD)
PB7	I2C SDA (LCD)

WIRING OVERVIEW



Efficient pin usage improves system design and reduces hardware complexity

WORKING PRINCIPLE

1

Initialization

System initializes sensors and LCD on power-up

3

ADC Conversion

STM32 ADC converts analog MQ9 signal to digital

5

Processing

STM32 processes and formats all sensor data

2

Gas Detection

MQ9 detects gas concentration via resistance change

4

Temp/Humidity

DHT11 sends 40-bit data frame over single wire

6

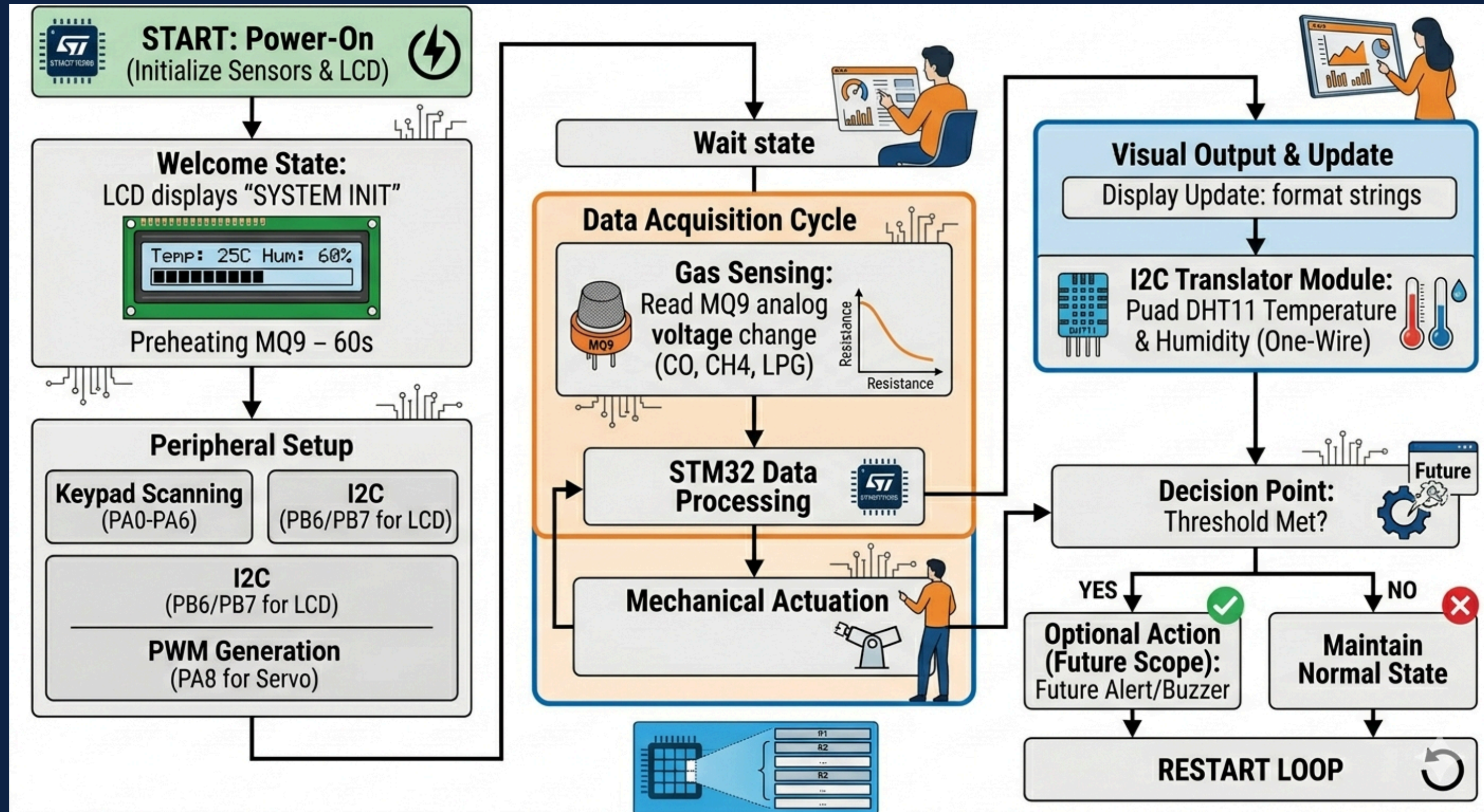
LCD Output

Data displayed on 16×2 I2C LCD in real-time

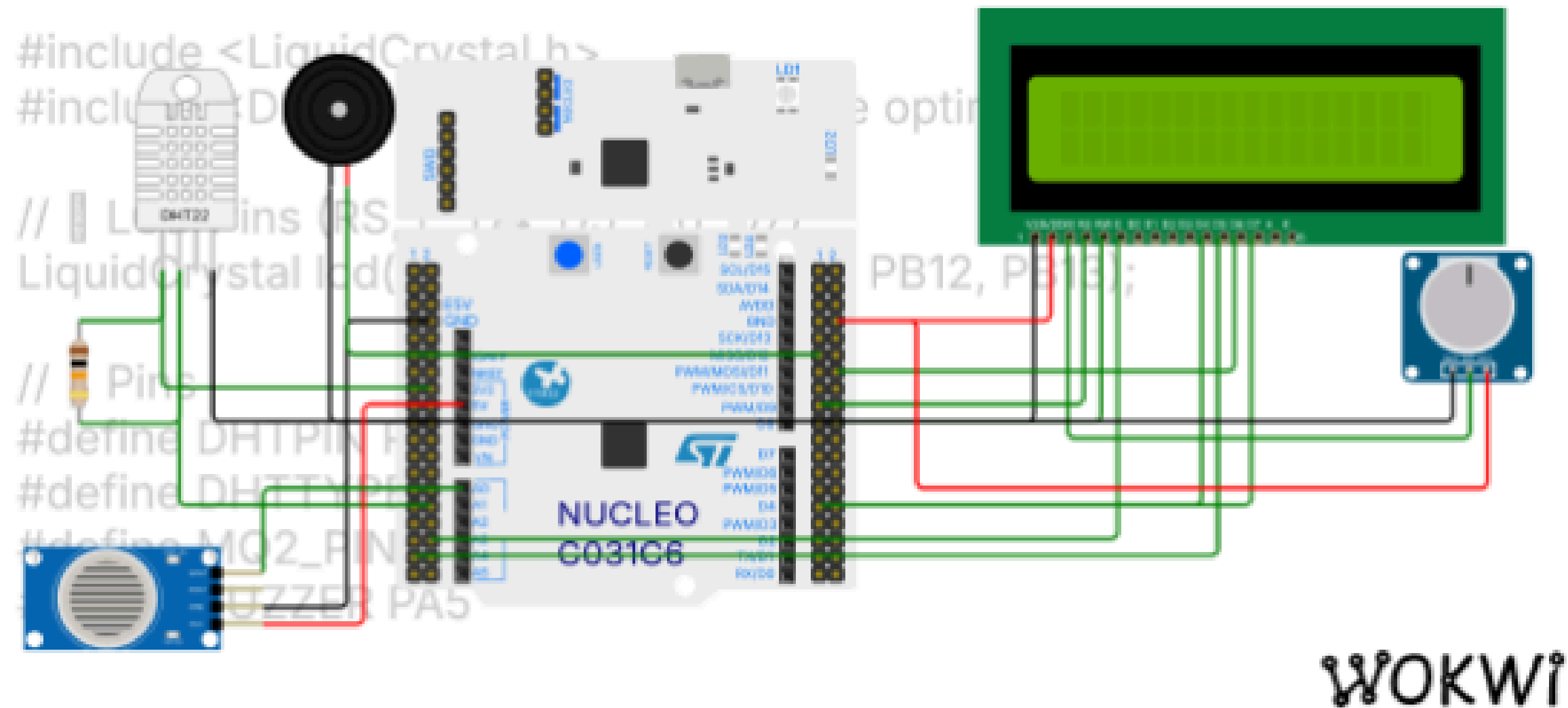
7

Loop: Cycle repeats continuously for real-time monitoring

Flowcharts



sketch.ino



WOKWI

Air Quality Monitoring System - Wokwi ESP32, STM32, Arduino Simulator

Run IoT and embedded projects in your browser: ESP32, STM32, Arduino, Pi Pico, and more. No installation required!

 WokwiMakes

CODES

```
for (j = 0; j < 8; j++) {

    // Wait for the HIGH pulse to start
    while (!HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_4));

    // Wait 40µs then sample the line
    delay_us(40);

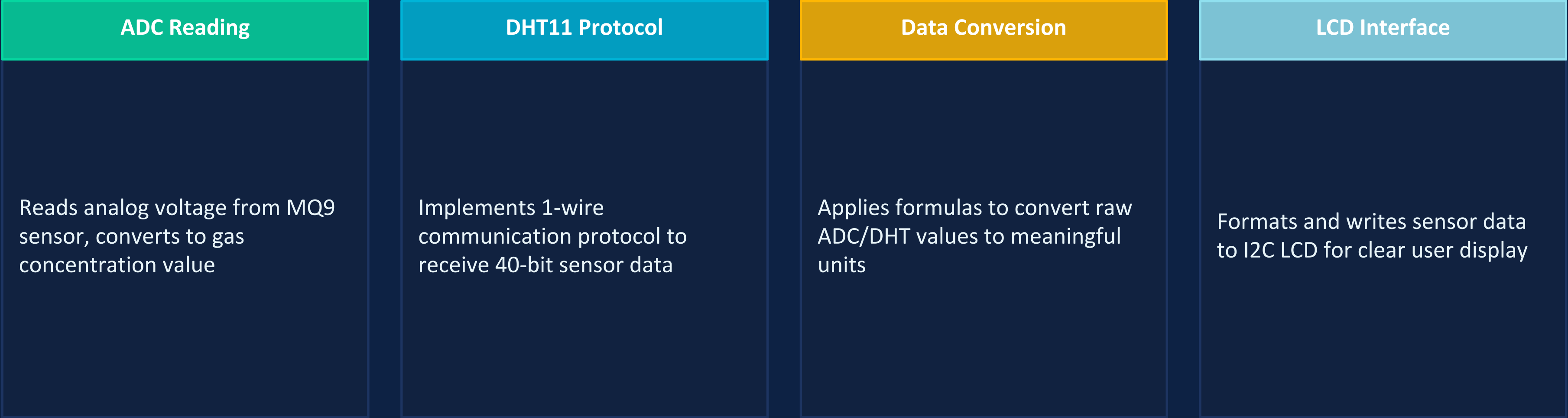
    if (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_4))
        byte[i] |= (1 << (7 - j)); // still HIGH = bit is 1

    // Wait for line to go LOW again (end of bit)
    while (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_4));
}
```

```
// mq9.c – read both outputs of the sensor
HAL_ADC_Start(hadc);
HAL_ADC_PollForConversion(hadc, 10);
data.analog_val = HAL_ADC_GetValue(hadc); // 0-4095
data.alarm = HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_1); // HIGH/LOW

// main.c – decide what to show on LCD
if (gas.alarm == GPIO_PIN_RESET) {
    lcd_send_string("GAS DETECTED!"); // alarm pin = LOW
} else {
    sprintf(buf, "Gas Lvl: %lu", gas.analog_val);
    lcd_send_string(buf); // show the number
}
```

SOFTWARE DESIGN & DATA PROCESSING



PROCESSING LOGIC



Optimized for real-time embedded performance

ADVANTAGES & LIMITATIONS

✓ ADVANTAGES

- ✓ Real-time environmental monitoring
- ✓ Low-cost and easy to build
- ✓ Portable and compact design
- ✓ Multi-parameter sensing in one device

✗ LIMITATIONS

- ✗ MQ9 requires periodic calibration
- ✗ DHT11 has limited accuracy range
- ✗ No buzzer/alert in the base design
- ✗ Not suitable for high-precision industrial use

APPLICATIONS & FUTURE SCOPE

Indoor Air Quality

Monitor homes, classrooms, and offices for CO & pollutants

Industrial Safety

Detect gas leaks in factories and chemical processing plants

Smart Buildings

Integrate with HVAC systems for automated environmental control

Env. Stations

Deploy in outdoor monitoring stations for city-wide data

FUTURE SCOPE

Add buzzer/alert for gas leakage detection

Wi-Fi / IoT integration for remote monitoring

☁ Cloud storage and real-time analytics dashboard

↑ Upgrade to MQ135 / DHT22 for higher accuracy

Battery-powered portable standalone device

Can evolve into a fully automated smart IoT-based environmental safety system

CONCLUSION

- Successfully implements a smart environmental monitoring system using STM32 microcontroller.
- Provides real-time data on gas concentration, temperature, and humidity, keeping users informed and safe.
- The system is cost-effective, scalable, and suitable for a variety of real-world applications.
- With further enhancements, it can become a fully automated smart monitoring and alert system for environmental safety.

Thank You